Track: Research Paper Hub

Principle: Transparency

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1. Milestone 1 at a Glance

Item	Details
Track	Track 3: Research Paper Discovery Hub (a track from the project list).
Principle	Transparency: users can see what the AI is, what data it uses, where answers come from, and what it cannot do.
Goal	Help students move from a general research interest to a specific research question, find relevant papers, understand the evidence, and explore possible research gaps.
AI tools used	ChatGPT (Thinking) for Concept A. Gemini (Pro) for Concept B.
Output 1	Two concept drafts side by side (Section 3).
Output 2	Comparison of strengths and weaknesses, and the chosen direction (Sections 4 and 5).
Output 3	Prompt log entries in the Appendix A (Section 6 and Appendices 1 and 2).
Output 4	Idea for approval (Section 2).

2. Idea for Approval

Field	Our proposal
Chatbot name	Research Paper Hub
Track and principle	Research Paper Discovery Hub / Transparency
Target users	University students in their 3rd or 4th year who are preparing their first research paper or graduation project and have not written a literature review before.
Problem it solves	Beginners cannot tell which AI statements about research are backed by real papers. They may trust invented citations, miss conflicting evidence, or treat AI suggestions as facts.
Principle feature in the AI pipeline (Transparency)	Every claim about the literature gets a label (Source-supported, AI interpretation, AI recommendation, or Unverified hypothesis) based on real pipeline checks. A citation check runs before display. The system says "I am not sure" when retrieval is weak.

Field	Our proposal
Why it is unique	A narrow user group, plus claim-level labels and a Show Evidence view that reflect what retrieval and citation checking actually did.

3. Two Concept Drafts Side by Side

Both tools received the same base prompt. Only the concept angle changed (the full texts are in Appendices 1 and 2).

	Concept A: Evidence-Guided Funnel (ChatGPT Thinking)	Concept B: Dialogic Evidence Explorer (Gemini Pro)
Idea	A guided, step-by-step funnel. The student approves each stage before moving on.	An open conversation. The student can jump between stages and ask for evidence on demand.
Number of stages	10 stages, from disclosure to a research brief.	5 stages (0 to 4), from onboarding to comparison and gaps.
Flow summary	Disclosure, interest, narrowing, question, paper search, paper analysis, comparison, gaps, review and approval, research brief.	Onboarding, topic and question, paper discovery, paper analysis, comparison and gap exploration.
Where transparency appears	Four-way claim labels, retrieved versus recommended papers, a rule for missing full text, and a transparency test plan.	Colored badges on claims and a Show Evidence drawer with the exact excerpt, author, and section.
Example conversation	7 turns with placeholder records (Example Record A, B, C).	6 turns on a 4-day workweek with placeholder papers and a conflicting-evidence badge.
Final output	A preliminary research brief with a verification checklist.	A literature synthesis outline. No defined brief or reading list.
Student control	Explicit, reversible approval at each stage.	Free navigation, but no explicit approval step.

Stages of each concept

Stage	Concept A: Evidence-Guided Funnel	Key transparency rule
1	Initial disclosure and onboarding	Disclosure must match the real corpus configuration.
2	Discover the student's interest	Topic suggestions are labeled as AI recommendations.
3	Narrow the interest	Feasibility advice is a recommendation, not proof.
4	Formulate a preliminary question	The question is a working draft the student approves and can revise.
5	Search for papers	Retrieved is not the same as recommended. No invented papers.
6	Analyze each paper	Author-reported, AI interpretation, and Not established are kept apart.
7	Compare papers	No averaging of incompatible scores. Conflicts are shown.
8	Explore research gaps	An AI-suggested gap is not an established gap.
9	Review, revise, approve	Approval is explicit and reversible.
10	Generate the research brief	Source IDs are kept. Verified details are separated from interpretations.

Stage	Concept B: Dialogic Evidence Explorer	Key transparency rule
0	Onboarding and AI disclosure	States the curated database, the limits, and the student's responsibility.
1	Topic exploration and question	Angles are labeled as AI-generated recommendations.
2	Paper discovery and selection	Papers come from the curated corpus only.
3	Paper analysis	Evidence-status labels on claims. Show Evidence opens the excerpt.
4	Comparison and gap exploration	Conflicting evidence is flagged. Gaps are labeled AI-generated hypotheses.

4. Strengths and Weaknesses

	Concept A: Evidence-Guided Funnel	Concept B: Dialogic Evidence Explorer
Strengths	• Strong student control: explicit approval at every stage. • Honest handling of missing full text and conflicting results. • Detailed error-handling table and a clear final brief.	• Easier to build: fewer stages. • Show Evidence is a concrete, testable interaction. • Flexible navigation fits how beginners explore.
Weaknesses	• Long: 10 stages are heavy for a 14-week build. • Evidence display is described only in general terms. • The example conversation is vague, with no concrete content.	• The example invents findings (a percentage and quoted passages), although the prompt forbade this. • Wipes context on topic change, so the student loses saved work. • Calls a research gap "strong" and "valid" too early, which weakens Transparency. • No final output and no explicit approval step. Conflict detection ("mathematically opposing findings") is unrealistic.

5. Chosen Direction

Decision

Concept A is stronger on student control, honest uncertainty, and a clear final output, but its 10 stages are heavy for a 14-week build. Concept B is easier to build and gives a concrete way to inspect evidence through its Show Evidence drawer, but it has no approval gates or final output, wipes the student's context when the topic changes, and endorses a research gap too early.**We chose a hybrid**: Concept A's staged flow and approval gates, reduced to 6 stages, combined with Concept B's on-demand Show Evidence drawer and per-claim labels. We chose it because it applies Transparency in the pipeline and in the interface, it is measurable, and its scope is realistic. We will build the citation check first.

Criterion	What we keep from A	What we keep from B
Flow	Approval gates and a final research brief	Free movement back to earlier stages
Transparency	Four-way claim labels and the retrieved versus recommended distinction	Show Evidence drawer with the exact excerpt
Feasibility	Error-handling table	Fewer stages and a simple interaction

Hybrid flow in 6 stages

#	Stage	Transparency in the AI pipeline	What the student sees	Student decision
1	Disclosure and onboarding	Disclosure is generated from the real corpus configuration.	What the AI is, the data it uses, its limits.	Start, ask about limits, or exit.
2	Interest and research question	Suggestions are tagged AI recommendation.	Draft question and open choices.	Approve, edit, or reject (revisable).
3	Paper discovery	Retrieval records why each paper matched. Abstains when retrieval is weak.	Shortlist with source IDs and match reasons.	Save or reject each paper.
4	Paper analysis	Claims are labeled. Citation check runs before display.	Method, findings, limits, and Show Evidence.	Mark relevant, uncertain, or irrelevant.
5	Comparison and gaps	Conflicts are detected across retrieved passages. Gaps are tagged Unverified hypothesis.	Comparison matrix and gap hypotheses.	Accept or reject each gap.
6	Review and research brief	Brief keeps source IDs and separates verified from interpreted.	Brief with a verification checklist.	Explicit approval, or hand over to a supervisor or librarian.

Show Evidence behavior

Evidence state	What Show Evidence does
Available	Shows the exact retrieved passage, the paper title, authors, year, and section where known, with a link to the source.
Incomplete	Says that only the abstract or metadata is available and limits the analysis to that material.
Conflicting	Shows the passages from different papers side by side and flags that they report different results on the same measure. Possible causes are labeled as hypotheses.
Unavailable	Says that no supporting passage could be found. The claim is shown as Unverified hypothesis or removed.

6. AI Prompt Log

Format follows Appendix A of the project file. The full prompt texts are in Appendices 1 and 2. Prompt 2 differs from Prompt 1 by more than the concept angle: it adds requirement 11 on Show Evidence and adjusts requirements 1, 3, and 6.

#	Tool	Exact prompt	What it gave	What we changed and why
1	Claude sonnet 5.5	Write one shared prompt for both tools	base prompt for both tools so the comparison is fair.	We made the prompt more precise and more relevant to the Transparency
2	ChatGPT (Thinking)	Concept A prompt: guided research funnel. See Appendix 1.	A 10-stage guided funnel with a fallback table, claim labels, and a 7-turn example.	Named a proposed corpus and a narrower user group. Reduced 10 stages to a 6-stage core for the 14-week limit. Added a concrete Show Evidence view, because the draft described evidence display only in general terms.

#	Tool	Exact prompt	What it gave	What we changed and why
3	Gemini (Pro)	Concept B prompt: open research conversation. See Appendix 2.	A 5-stage open conversation with badges, a Show Evidence drawer, and fallback messages.	Removed the invented findings (a percentage and quoted passages), because our prompt forbade fabricated findings. Replaced the context wipe with saved, reopenable work. Replaced the AI's praise of a gap with the neutral label Unverified hypothesis. Added an approval step and a final output. Replaced "mathematically opposing findings" with a realistic rule: flag a conflict when retrieved passages from different papers report different results on the same measure.

Appendix 1: Exact Prompt for Concept A (ChatGPT Thinking)

Act as an AI Product Architect specializing in transparent AI-powered academic research assistance.
Project: Research Paper Hub, an AI Research Mentor for undergraduate university students beginning their first research paper or graduation project.
Track: Transparency.
Target users: Undergraduate students with limited experience in academic research and literature reviews.
Grounding source: Assume the team will use a curated collection of academic papers relevant to undergraduate research. The exact corpus, research field, and size have not yet been finalized. Clearly identify these as design assumptions and do not invent a finalized dataset.
Concept angle: Design this as a guided, step-by-step research funnel. The chatbot leads students through clearly defined stages, asks focused questions, and helps them complete one stage before moving to the next. Students can revisit earlier stages or change their research direction when necessary.
Objective
Design the chatbot's core interaction flow to guide students from an initial research interest to a preliminary research question, relevant academic papers, paper analysis, paper comparison, and potential research gap exploration.
Requirements
1. Define the main stages of the interaction from the student's first message to the final output.
2. For each stage, describe:
- Student's goal.
- Example student input.
- AI response.
- Decisions or choices available to the student.
- Resulting output.
3. Include alternative paths for unclear research interests, changing research topics, insufficient evidence, conflicting findings, and requests to revisit earlier stages.
4. Explain how the chatbot helps students discover academic papers and understand each paper's methodology, findings, and limitations.
5. Show how students compare papers and explore potential research gaps without treating AI-generated hypotheses as established facts.
6. Embed transparency throughout the flow by distinguishing source-supported claims from AI-generated interpretations and recommendations, providing traceable citations where available, communicating uncertainty, and allowing students to verify important claims.
7. Keep students responsible for reviewing and approving their research questions, paper selections, and conclusions. The chatbot must not make these decisions on their behalf.
8. Begin with an AI disclosure that explains what the chatbot is, what research collection it uses, what its limitations are, and why students should verify important claims.
9. Handle out-of-scope questions, weak retrieval, missing citations, API errors, and time-outs with clear fallback messages. Include a handover or exit option, such as consulting a supervisor or academic librarian.
10. Explain which transparency mechanisms operate inside the AI pipeline, including retrieval, source-grounded generation, citation validation, claim labeling, and handling unsupported claims, rather than describing only what appears on screen. Present these as proposed capabilities, not features that are already implemented.
11. Keep the design realistic for a 14-week student team project. Focus on interaction logic and system behavior, not visual UI design or implementation code.
Deliverables

A. **Concept name and rationale:** Give the concept a clear name and briefly explain why a guided funnel suits beginner researchers.
B. **Complete numbered interaction flow:** Describe the full journey, including normal paths, alternative paths, error handling, and the final output.
C. **Realistic example conversation:** Create a fictional student interaction covering the initial interest, refinement of the research question, paper discovery, paper analysis, comparison, and possible research gaps. Clearly label invented conversational examples. Do not fabricate real academic papers or citations.
D. **Transparency mechanisms:** Explain the mechanisms used at each stage, separating internal AI pipeline controls from information shown to the student.
E. **Strengths and weaknesses:** Provide exactly three strengths and three weaknesses of this concept.
Constraints
- Do not invent academic sources, paper titles, DOIs, citations, or research findings.
- Do not claim that proposed capabilities are already implemented.
- Clearly distinguish retrieved evidence, AI-generated interpretations, recommendations, and unverified hypotheses.
- If evidence is insufficient, state this explicitly instead of generating unsupported conclusions.
- Keep the output structured, specific, and suitable for a university Human-AI Interaction course milestone report.
- Use numbered steps, concise explanations, and realistic examples.
- Do not write the final comparison with another concept or select a winning concept. Produce this concept draft independently.

Appendix 2: Exact Prompt for Concept B (Gemini Pro)

Prompt 2: Concept B | Open Research Conversation

Act as an AI Product Architect specializing in transparent AI-powered academic research assistance.
Project: Research Paper Hub, an AI Research Mentor for undergraduate university students beginning their first research paper or graduation project.
Track: Transparency.
Target users: Undergraduate students with limited experience in academic research and literature reviews.
Grounding source: Assume the team will use a curated collection of academic papers relevant to undergraduate research. The exact corpus, research field, and size have not yet been finalized. Clearly identify these as design assumptions and do not invent a finalized dataset.
Concept angle: Design this as an open research conversation. Students can explore their interests, ask follow-up questions, jump between research stages, and change topics freely. An on-demand "Show Evidence" feature lets students inspect supporting paper excerpts and citations. Every substantive claim about the literature receives a visible evidence-status label, such as "Supported by retrieved sources," "Partially supported," "Conflicting evidence," or "Not verified." Confidence labels must not be presented as proof of factual correctness.
Objective
Design the chatbot's core interaction flow to guide students from an initial research interest to a preliminary research question, relevant academic papers, paper analysis, paper comparison, and potential research gap exploration.
Requirements
1. Define the main stages of the interaction from the student's first message to the final output, while allowing students to move between stages freely.
2. For each stage, describe:
- Student's goal.
- Example student input.
- AI response.
- Decisions or choices available to the student.
- Resulting output.
3. Include alternative paths for unclear research interests, changing research topics, insufficient evidence, conflicting findings, and requests to revisit earlier discussions.
4. Explain how the chatbot helps students discover academic papers and understand each paper's methodology, findings, and limitations.
5. Show how students compare papers and explore potential research gaps without treating AI-generated hypotheses as established facts.
6. Embed transparency throughout the flow by distinguishing source-supported claims from AI-generated interpretations and recommendations, providing traceable citations where available, communicating uncertainty, and allowing students to verify important claims through the "Show Evidence" feature.
7. Keep students responsible for reviewing and approving their research questions, paper selections, and conclusions. The chatbot must not make these decisions on their behalf.
8. Begin with an AI disclosure that explains what the chatbot is, what research collection it uses, what its limitations are, and why students should verify important claims.
9. Handle out-of-scope questions, weak retrieval, missing citations, API errors, and time-outs with clear fallback messages. Include a handover or exit option, such as consulting a supervisor or academic librarian.

10. Explain which transparency mechanisms operate inside the AI pipeline, including retrieval, source-grounded generation, citation validation, claim labeling, and handling unsupported claims, rather than describing only what appears on screen. Present these as proposed capabilities, not features that are already implemented.
11. Explain how the "Show Evidence" feature behaves when evidence is available, incomplete, conflicting, or unavailable. Describe how claim labels are assigned and their limitations. Do not imply that a confidence score alone guarantees correctness.
12. Keep the design realistic for a 14-week student team project. Focus on interaction logic and system behavior, not visual UI design or implementation code.

Deliverables

A. Concept name and rationale: Give the concept a clear name and briefly explain why open conversation and on-demand evidence access suit beginner researchers.
B. Complete numbered interaction flow: Describe the full journey, including flexible navigation, alternative paths, error handling, and the final output.
C. Realistic example conversation: Create a fictional student interaction covering the initial interest, refinement of the research question, paper discovery, paper analysis, comparison, and possible research gaps. Include at least one use of "Show Evidence" and one example of uncertain or conflicting evidence. Clearly label invented conversational examples. Do not fabricate real academic papers or citations.
D. Transparency mechanisms: Explain the mechanisms used at each stage, separating internal AI pipeline controls from information shown to the student.
E. Strengths and weaknesses: Provide exactly three strengths and three weaknesses of this concept.

Constraints

- Do not invent academic sources, paper titles, DOIs, citations, or research findings.
- Do not claim that proposed capabilities are already implemented.
- Clearly distinguish retrieved evidence, AI-generated interpretations, recommendations, and unverified hypotheses.
- If evidence is insufficient, state this explicitly instead of generating unsupported conclusions.
- Keep the output structured, specific, and suitable for a university Human-AI Interaction course milestone report.
- Use numbered steps, concise explanations, and realistic examples.
- Do not write the final comparison with another concept or select a winning concept. Produce this concept draft independently.

Claude Prompt : Write one shared prompt for both AI tools that describe that can fit this milestone

1. Two concept drafts side by side. Use two different AI tools to draft two concepts for the chatbot’s core interaction flow.
2. A short comparison of the strengths and weaknesses of the two drafts, and which direction they chose.
3. Prompt log entries. These are the exact prompts, the tool name, what it gave, and what the team changed and why (Appendix A).
4. The idea for approval. It must be unique < since two teams can share a track but not the same user group and idea>